GCSE · OCR · Computer Science

2 hours 27 questions

Primary & Secondary Storage

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The Need for Primary Storage / RAM & ROM / Virtual Memory / The Need for Secondary Storage / Common Types of Storage / Suitable Storage Devices

Easy (7 questions)	/20
Medium (12 questions)	/46
Hard (8 questions)	/43
Total Marks	**/109**

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Easy Questions

1 (a) Nina wants to transfer photos from a digital camera to an external **secondary storage** device.

Identify the **three** common **types of storage** Nina can choose from.

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..... **(3 marks)**

(b) **State four characteristics** of **secondary storage devices** that Nina should consider when choosing a device.

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..... **(4 marks)**

2 Ali's tablet computer has ROM (read only memory).

Describe the **purpose of ROM** in Ali's tablet computer.

.....

..... **(2 marks)**

3 (a) Both computers have **RAM** and **ROM**.

The table has five statements describing RAM and/or ROM.

Tick (✓) one or more boxes in each row to identify if that statement describes RAM and/or ROM.

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	RAM	ROM
Stores data		
The memory is volatile		
Data will not be lost when the computer is turned off		
Data is read-only, cannot be changed		
Stores currently running data and instructions		

(5 marks)

(b) **Give one difference** between **RAM** and **flash memory**.

..... **(1 mark)**

4 A musician is trying to decide if they should buy a magnetic secondary storage device or a solid state secondary storage device for their computer

Identify **one other** type of secondary storage.

..... **(1 mark)**

5 **Identify** the two main types of primary storage found in a typical general-purpose computer system

.....

..... **(2 marks)**

6 **State** which type of primary storage is **volatile** and will lose its contents when the power is turned off

..... **(1 mark)**

7 **Identify** which type of primary storage is used to store the **start-up instructions** (bootstrap) for a computer

..... **(1 mark)**

Medium Questions

1 (a) A smart television allows the user to search the Internet and watch videos online.

The smart television has both RAM and ROM.

State the **difference** between **RAM** and **ROM**.

..... **(1 mark)**

(b) **Give two examples** of **data** that the smart television could **store** in **RAM**.

.....

..... **(2 marks)**

2 (a) The smart television has **secondary storage**.

State, using **an example**, why the smart television needs **secondary storage**.

.....

..... **(2 marks)**

(b) **Identify** one appropriate **type of secondary storage** for the smart television. **Justify** your choice.

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.....

..... **(4 marks)**

3 Nina wants to transfer photos from a digital camera to an external secondary storage device.

Define what is meant by '**secondary storage**'.

..... **(1 mark)**

4 **Tick (✓) one** box in **each row** to show whether each of the statements is **true** for the RAM or ROM in a Sat Nav.

	RAM	**ROM**
Stores the boot up sequence of the Sat Nav.		
The contents are lost when the Sat Nav is turned off.		
Holds copies of open maps and routes.		

(3 marks)

5 A computer system has a 2.5 GHz processor and 5 GB of RAM.

Complete the paragraph about memory by **writing the missing terms** in the spaces, **using the words provided**.

cache	**data**	**hard drive**	**write**
instructions	**changed**	**closing**	**memory stick**
operator	**primary**	**random**	**virtual**
read	**secondary**	**start-up**	**translator**

ROM stands for ..... only memory.

This stores the ..... instructions for a computer and cannot be ..... .

RAM stands for ..... access memory. This stores the instructions and ..... that are currently being used. If the computer does not have enough RAM to run a process it can make use of ..... memory.

RAM and ROM are both examples of .....memory.

Memory located close to the processor that allows faster access than from RAM is called .....memory.

(8 marks)

6 (a) Charlie has purchased a new tablet computer. The tablet has an **internal secondary storage** device.

Describe what the internal secondary storage device **will store**.

.....

..... **(2 marks)**

(b) The storage device is a solid state device.

Give three benefits of the tablet having a **solid state device** instead of a **magnetic device**.

.....

.....

..... **(3 marks)**

(c) **Give two drawbacks** of the tablet having a **solid state device** instead of a **magnetic device**.

.....

..... **(2 marks)**

7 A musician uses a computer to make and record music

The musician has run out of storage space on their secondary storage device and needs to buy a replacement.

Identify whether the musician should buy a magnetic secondary storage device or a solid state secondary storage device for their computer.

Justify your choice.

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.....

.....

..... **(4 marks)**

8 (a) A car has a 'Follow Me' system that uses a cruise control feature to allow the car to follow the car in front of it. It will keep the same speed and distance without the driver's intervention. The cruise control system is an example of an embedded system.

The car's system has Read Only Memory (ROM) and Random Access Memory (RAM).

State **two** items that will be stored in the ROM for the 'Follow Me' system.

.....

..... **(2 marks)**

(b) The RAM will store currently running data and instructions.

State **three** items of data that will be stored in the RAM for the 'Follow Me' system.

.....

.....

..... **(3 marks)**

(c) **Explain** why the 'Follow Me' system **does not** need virtual memory.

.....

..... **(2 marks)**

9 **Describe** the purpose of RAM in a computer system and why it is needed for a computer to function

.....

..... **(2 marks)**

10 **Explain** why a computer requires **ROM** in addition to RAM, specifically referring to what happens when the computer is first switched on

.....

..... **(2 marks)**

11 A student is using a computer for photo editing.

Describe how the **Operating System** uses primary storage to allow the student to perform this task

.....

..... **(2 marks)**

12 **Identify** one benefit of a computer system having a **large capacity of RAM** when a user is running multiple applications at the same time

..... **(1 mark)**

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Hard Questions

1 (a) Kerry has 5GB of files to transfer from her laptop at work to her new computer. She has been told to buy an external solid state device to do this.

Give one example of a **solid state device**.

..... **(1 mark)**

(b) **Identify** whether the device given in **part (a)** is an example of **primary** or **secondary** memory.

..... **(1 mark)**

(c) Kerry was originally going to use an optical storage device to transfer her files.

Discuss whether an **optical** or **solid state device** is the **most appropriate media** to **transfer these files**.

You may want to consider the following characteristics in your answer:

- **portability**
- **robustness**
- **capacity**
- **cost**

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..... **(8 marks)**

(d) The filesizes of Kerry's files are usually displayed in **megabytes** (MB) or **gigabytes** (GB).

Calculate how many MB are in 5GB. **Show your working**.

.....

..... **(2 marks)**

2 (a) William is creating a film for a school project using a digital video camera.

The digital video camera has a secondary storage device.

Explain why the digital video camera **needs secondary storage**.

.....

..... **(2 marks)**

(b) The digital video camera uses solid state storage.

Explain why solid state storage is the **most appropriate type of storage** for the digital video camera.

.....

.....

.....

..... **(4 marks)**

3 (a) Alicia's computer only has **2GB of RAM**, but she says that **virtual memory** can be used instead of adding more RAM.

Explain how virtual memory can **compensate for the lack of RAM** in Alicia's computer.

.....

.....

..... **(3 marks)**

(b) **Explain** why it would be **beneficial for Alicia** to **get more RAM** instead of relying on virtual memory.

.....

..... **(2 marks)**

4 A user is running several memory-intensive programs, and the RAM has become full.

Explain how **virtual memory** will be used to allow the computer to continue functioning

.....

.....

..... **(3 marks)**

5 A specialised cruise control system in a car is an example of an **embedded system**.

Explain why this specific system is unlikely to require **virtual memory**, even if its RAM capacity is low

.....

..... **(2 marks)**

6 **Compare** primary storage with secondary storage, focusing on the **need** for each in a computer system and their different characteristics regarding **volatility** and **access speed**

.....

.....

..... **(3 marks)**

7 A small photography business needs to store and transport large image files between their studio and client locations. They are considering using either an external hard disk drive (HDD) or a USB flash drive.

Evaluate which type of secondary storage device would be most appropriate for this scenario, considering factors such as capacity, portability, durability, and cost.

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..... **(6 marks)**

8 A local veterinary clinic needs to back up patient records and medical imaging files. The clinic manager is deciding between using a solid-state drive (SSD) or a magnetic hard drive (HDDs) for this purpose.

Compare these two types of secondary storage devices for this use case.

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..... **(6 marks)**